

The Pleistocene-Holocene Transition in Tagua-Tagua, central Chile

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Abstract

Paleohydrological reconstructions from the Chilean Altiplano document abrupt moisture fluctuations during the last millennia. Although the end of the mid Holocene aridity and the onset of more humid conditions between 6–4 ka has been identified in numerous regional marine and terrestrial sites, the timing of late Holocene dry and humid phases shows large regional variability. Laguna Miscanti and Laguna Miñiques (23° 43'S – 67° 46'W, 4140 m asl) are topographically closed, but connected by surface outflow from Miscanti. Sedimentological and geochemical indicators from two new cores show large facies changes, i.e. higher carbonate and evaporite deposition during more arid periods and increased organic productivity (both algal and macrophyte) during more humid phases. As in most Andean lakes located in volcanic settings, large 14C reservoir effects occur complicating 14C dating, so the age models include 210Pb and U/Th dating. In spite of dating uncertainties, both lakes show similar patterns during the last millennium. A humid phase in Laguna Miscanti prior to ca 1200 CE is coherent with rodent middens and geomorphological features indicative of a major pluvial/recharge event at lower altitudes (Atacama Desert) during the Medieval Climate Anomaly (ca 800 - 1300 CE). The LIA (1300 – 1850 CE) is characterized by several arid/humid cycles and the last century by a productivity increase. The hydrological changes observed during the last millennium illustrate the complex dynamics of recent climate evolution over the high altitude Andean plateau. Discrepancies between the northern and southern Altiplano records and with intermediate latitudes (Central Chile) records may reflect contrasting responses to external forcing (Westerlies versus South American Monsoon dynamics) along different climatic zones.

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1. 1. Introduction

2. 2. Study site and regional setting

3. 3. Materials and methods

In November 2019, we recovered a sedimentary sequence of 15 cm length, 10 cm width and 280 cm depth of the profile exposed on the western wall of a 1x2 m excavation in the ALTT (Tagua-Tagua 3, Figure 1). Sediment samples (5 to 10 g of material) were obtained at 1-cm intervals. Sixty-six samples were sent to the laboratories of Pyrenees Ecology Institute (IPE-CSIC) and the Andalusian Institute Earth Sciences Institute (IACT-CSIC) in Spain for geochemical, pollen and isotopic analysis. Another set of samples were sent to the Archaeobotany and Palaeoecology Laboratory of the University of Exeter for the processing and extraction of diatom and phytolith. Sedimentary facies were defined and characterized based on macroscopic description (color, bedding features, grain size, lithology and sedimentary textures) and a machine learning workflow using the scikit-learn library (cite) in Python. Sedimentary facies were defined using a dimensionality reduction (Exploratory Factor Analysis) and unsupervised machine learning (K-Means algorithm cluster) with three different dataset (elemental geochemistry, grain size and diatoms)(Hall., 2016).

Samples were taken every 2 cm for analysed of total carbon (TC), total inorganic carbon (TIC), total organic carbon ($TOC = T - TIC$), total nitrogen (TN) and total sulfur (TS) in a LECO SCN furnace (IPE-CSIC). The grain-size frequency distributions (between 0.02 μ m and 2000 μ m) were measured in triplicate using a Malvern Mastersizer 2000 APA2000 grain-size analyzer with Hydro 2000G AWA2000. To determine the grain-size, the samples were pre-treated with 10% H₂O₂ and 10% HCl to remove organic matter and carbonates, respectively. Robust End-member mixing analysis (EMMA) was performed using the generic deterministic EMMA protocol with the R package EMMAgeo v0.9.8. (Dietze et al 2012; Dietze and Dietze 2019) to infer the sediment sources, transport and depositional processes. Stable carbon and oxygen isotope analyses on calcite were carried out every 2 cm by mass spectrometry in bulk sediment samples following standard procedures and using an IRMS Finnigan MAT 251. The isotopic values are reported in the conventional delta notation relative to the PDB standard.

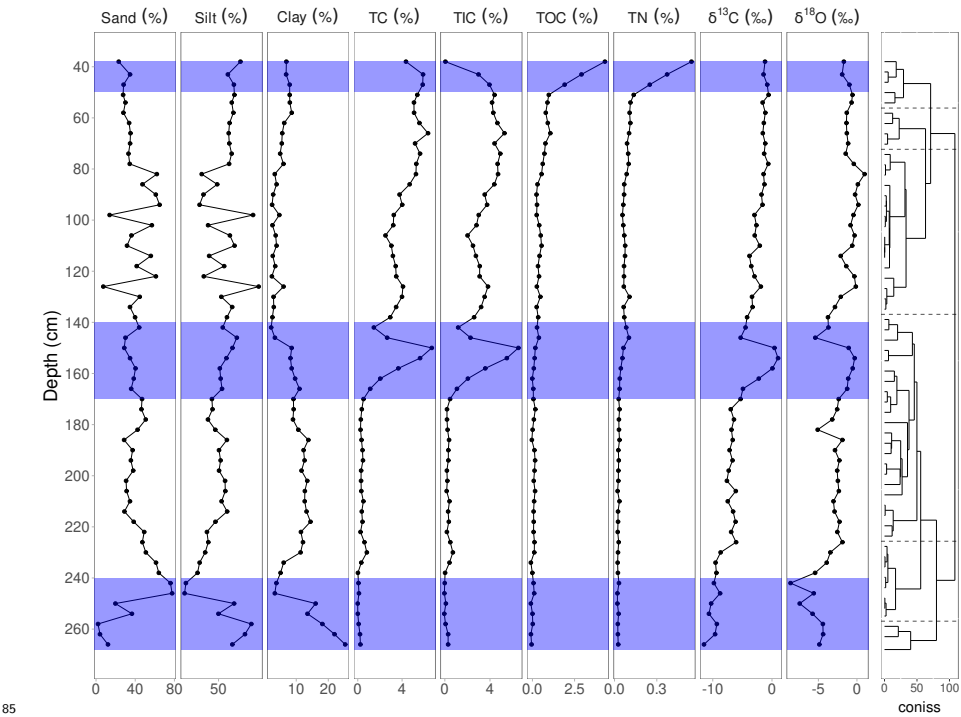
Eleven pollen samples were processed following Faegri and Iversen (1989) methods to extract pollen grains. Lycopodium tablets were added for calculating pollen concentration (grains/cm³; Stockmarr(1971)) and accumulation rates (grains/cm²/yr). Pollen grains were mounted on glass slides and identified and quantified under an Axiostar Carl Zeiss microscope (400x and 1000x) using published pollen atlas Heusser (1971) and the reference pollen collection of the Paleocology Laboratory at IPE-CSIC. Pollen counts include 300 terrestrial pollen grains excluding paludal taxa. Phytoliths were obtained from twenty four samples of approximately 2 gr of sediment. We followed the standard procedures for phytoliths extraction (Piperno, 1988, 1994, 2006) using 36% HCl and 70% HNO₃ to remove carbonates and organic matter respectively.

60 Phytoliths were separated by heavy liquid flotation using ZnBr₂ prepared to a
61 density of 2.3 g/cm³ (Dickau et al., 2013). Finally, phytoliths were counted until
62 obtaining a total of 200 per sample and they were classified according to the
63 International Code for Phytolith Nomenclature (Madella et al. 2005). Diatoms
64 were extracted simultaneously with phytoliths. Digested aliquotes were mounted
65 onto permanent slides with Naphrax. Diatom valves were visualized under a
66 XXXXX microscope, with an oil immersion objective of 1000x. A minimum of
67 400 valves were counted for each level, with the exception of the base levels X,
68 Y, Z, where valves were scarce. Diatoms were identified to species level with
69 the LM images (1000x) taken using a digital SLR camera (Canon EOS Rebel)
70 attached to microscope AND SEM???. The following literature was used for
71 taxonomy and autoecology of the encountered valves.

72 The TT3 chronology was reconstructed with thirteen AMS-radiocarbon
73 dates on charcoal and bulk sediment determined at DirectAMS laboratories,
74 USA. Results are presented in units of percent modern carbon (pMC) and the
75 uncalibrated radiocarbon age before present (relative to 1950 CE). Radiocarbon
76 ages were calibrated using the Southern Hemisphere calibration curve (SHCal20,
77 Hogg et al 2020). We used the package rbacon v2.2 (Blaauw and Christen, 2011)
78 for age–depth modeling and to establish the deposition rates along the sequence.
79 A change point analysis was carried out to identify significant breaks in the time
80 series (Bai and Perron, 2003; Killick and Eckley, 2014). Stratigraphic diagrams
81 were analyzed and plotted using ‘tidypaleo’ v0.1.1, ‘ggplot’ v3.3.5 and ‘rioja’
82 v0.9-26 packets in R 4.1.0 software.

83 **4. Results**

84 *4.1. Cores correlations and sedimentary proxies*



86 **References**